

HygienePrio: Cyber-Hygiene Signal Augmentation for EPSS-Weighted Vulnerability Prioritization

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Abstract: Capacity-constrained vulnerability remediation teams must decide which (host, CVE) pairs to patch within each maintenance window, yet the dominant public triage signal, the Exploit Prediction Scoring System (EPSS), is asset-agnostic by design: it estimates per-CVE exploit likelihood without regard to whether the affected host is actively managed, carries high patch debt, or is exposed through privileged Active Directory (AD) relationships. The missing signal is therefore dynamic, host-level hygiene posture rather than additional CVE-level features.

We present HygienePrio, a scoring framework that integrates a three-dimensional Hygiene Risk Score (HRS), capturing patch posture, AD exposure state, and telemetry freshness, into an EPSS-weighted (host, CVE) prioritization scorer with an explicit interaction term, calibrated by pre-registered grid search on held-out fleet seeds. To our knowledge, HygienePrio is the first scorer to jointly operationalize per-CVE exploit likelihood and host-level hygiene posture for remediation ordering within a pre-registered, fully reproducible evaluation framework.

On a pre-registered, reproducible benchmark across 25 independent fleet seeds, HygienePrio raises mean Precision@50 from EPSS-only's 0.202 to 0.509 - an approximately 31-percentage-point gain under capacity-constrained schedules, with non-overlapping BCa 95% confidence intervals and an improvement on every one of the 25 seeds. We further validate the scorer against *real public EPSS and Known-Exploited-Vulnerability (KEV) attribute distributions* drawn from a 203,174-CVE snapshot: there the operationally significant advantage - host-level discrimination at major-CVE disclosure moments, exactly when remediation capacity is scarcest - persists, and we characterize the regimes in which hygiene augmentation helps and those in which it is dominated by capacity. An ablation isolates patch posture as the load-bearing hygiene dimension (≈ 20 pp at P@50), well above AD exposure (≈ 9 pp) and telemetry freshness (≈ 2 pp). The framework, the benchmark generator, and the frozen results are released so that every number reproduces deterministically from fixed seeds.

Index Terms: vulnerability prioritization, EPSS, cyber-hygiene, patch posture, Active Directory, telemetry freshness, remediation scheduling, reproducible benchmark, reproducibility

I. INTRODUCTION

Public-sector and enterprise endpoint fleets accumulate disclosed vulnerabilities faster than operations teams can remediate them within constrained maintenance windows. The practical triage question is not *which vulnerabilities are inherently severe* but *which (host, CVE) pairs should receive the next unit of limited remediation capacity*, a distinction that requires combining exploit-likelihood signals with knowledge of which hosts are most exposed and least well-maintained.

The Exploit Prediction Scoring System (EPSS) [1] has emerged as the leading public signal for per-CVE exploit likelihood, outperforming CVSS base scores as a predictor of

near-term exploitation in the wild [2]. EPSS is incorporated into risk-based vulnerability management (RBVM) tools, CISA vulnerability prioritization guidance [3], [4], and remediation policy frameworks. However, EPSS is asset-agnostic by design: it assigns a single probability score to a CVE regardless of whether the vulnerable host is a freshly-telemetered, heavily-patched workstation or an unmanaged endpoint with stale patch data and over-privileged AD group memberships.

Intuitively, a CVE with moderate EPSS on a host that has gone 21 days without telemetry, carries 40% unpatched CVE debt, and is accessible from domain-privileged accounts represents a materially different risk than the same CVE on a well-maintained host with fresh telemetry and minimal AD exposure. Yet EPSS rankings treat both identically. Host-level *hygiene posture*, the aggregate state of patch compliance, identity and access management hygiene, and telemetry coverage, is a natural complement to exploit-likelihood scoring.

Prior work directly relevant to this problem is **VulnPrio** [5] (referred to throughout as the VulnPrio study), which built a reproducible benchmark for (host, CVE) pair prioritization on a synthetic government-shaped fleet (EEHDA) and evaluated 13 strategies combining EPSS, CVSS, KEV membership, asset criticality, and endpoint exposure signals. The VulnPrio study's central null finding was that the proposed multi-factor model was statistically indistinguishable from the EPSS-only baseline: inter-strategy differences fell within seed-to-seed variation, and no CVE-level augmentation strategy materially outperformed EPSS. The VulnPrio study concluded that this result exposed capacity-driven and base-rate trade-offs and provided a falsifiable benchmark on which calibrated studies could build, explicitly framing the result as a foundation for future work rather than an endpoint.

Separately, **HygieneBench** [6] introduced a reproducible benchmark for cyber-hygiene anomaly detection on a synthetic fleet covering Active Directory identity state, endpoint patch posture, vulnerability exposure, and telemetry freshness. HygieneBench's primary finding was that 86.2% of evaluated (condition, task, method) configurations fail to outperform a rule baseline, but that patch-vulnerability hygiene (Task T5, +0.210 average precision over rule for OCSVM) and group-membership drift (Task T2, +0.185 AP for temporal z-score) are exceptions where structured ML methods add meaningful signal.

The central hypothesis of this paper is that the signal missing from the VulnPrio study's multi-factor model was not additional CVE-level features but host-level hygiene posture: patch debt, AD privilege exposure, and telemetry fresh-

ness. HygieneBench demonstrated that these dimensions carry discriminative structure detectable from the synthetic fleet telemetry. HygienePrio operationalizes this hypothesis by constructing a Hygiene Risk Score (HRS) from HygieneBench-style host telemetry and integrating it into an EPSS-weighted (host, CVE) scorer via an additive combination with an explicit interaction term.

A. Contributions

This paper makes the following contributions:

- **C1, HygienePrio scorer:** A four-component weighted scoring function $S(h, c) = \alpha \times \text{EPSS}(c) + \beta \times \text{HRS}(h) + \gamma \times \text{KEV_recency}(c) + \delta \times (\text{EPSS}(c) \times \text{HRS}(h))$ that integrates host-level hygiene posture with per-CVE exploit likelihood in a principled, interpretable form.
- **C2, Hygiene Risk Score (HRS):** A normalized, three-dimensional host-level hygiene signal covering patch posture, AD exposure state, and telemetry freshness, derived from the HygieneBench fleet schema and calibrated on held-out seeds.
- **C3, Empirical evaluation:** A pre-registered, fully reproducible 25-seed benchmark demonstrating that HygienePrio achieves approximately 31 pp improvement in Precision@50 over EPSS-only (0.509 vs. 0.202; non-overlapping BCa 95% CIs), with patch posture as the dominant HRS dimension (~20 pp ablation drop) and AD exposure as a substantial secondary contributor (~9 pp).
- **C4, Failure mode analysis and operationalization boundary:** An explicit characterization of the conditions under which HygienePrio's advantage is smaller (interaction term equivocal; freshness dimension marginal) and the boundary conditions (ground-truth circularity, generated CVE attributes, 5-seed calibration) that delimit the positive finding, situating the result as a validated benchmark contribution ahead of prospective real-fleet deployment.
- **C5, External validity and adversarial robustness:** A partial external-validity check that re-evaluates the scorer under real EPSS/KEV attribute distributions from frozen public feeds, showing the aggregate Precision@50 gap shrinks from +31 pp to +2.4 pp, and a Stackelberg best-response analysis showing HygienePrio degrades by ≤ 3 pp under EPSS-gaming attacks that cost EPSS-only ≈ 14 pp.
- **C6, Retrospective and information-theoretic framing:** A worked retrospective on ten high-profile real CVEs identifying host-side discrimination at major-CVE disclosure moments as the operationally significant advantage, and a mutual-information bound argument delimiting how much hygiene augmentation could possibly help on real data.

B. Paper Organization

Section 2 reviews background. Section 3 formalizes the problem. Section 4 describes the benchmark dataset and fleet. Section 5 presents the HygienePrio methodology. Section 6 describes the experimental setup. Section VII reports the

primary benchmark results. Section 8 discusses findings, limitations, and connections to prior papers. Section IX addresses threats to validity. Section X reports the external-validity evaluation under real EPSS/KEV attribute distributions. Section XI analyzes adversarial robustness under a Stackelberg attacker. Section XII presents a retrospective on ten high-profile real CVEs. Section XIII develops an information-theoretic ceiling argument. Section 14 surveys related work, and Section 15 concludes.

II. BACKGROUND AND RELATED WORK

A. EPSS and Exploit-Likelihood-Based Prioritization

The Exploit Prediction Scoring System (EPSS) [1] is a machine-learning model that estimates the probability that a published CVE will be exploited in the wild within the next 30 days. EPSS v3 uses $\sim 1,500$ features drawn from CVE descriptors, NVD metadata, and threat intelligence feeds, and produces daily probability scores for all published CVEs. Empirical evaluations [2] have shown EPSS substantially outperforms CVSS base scores as a predictor of near-term exploitation: CVEs in the top decile of EPSS account for a disproportionate fraction of observed exploitation events. EPSS is now integrated into CISA's vulnerability prioritization guidance and is used by numerous RBVM vendors.

CISA's Known Exploited Vulnerabilities (KEV) catalog [7] is a complementary signal: it lists CVEs confirmed to have been exploited in the wild, with mandated remediation deadlines for federal agencies under BOD 22-01 [3]. KEV membership provides strong prioritization signal but covers only a small fraction of the CVE backlog ($\sim 15,000$ of the 200,000+ published CVEs as of 2026). Prior work has studied optimal combined use of EPSS and KEV [5], generally finding that KEV provides a high-precision but low-recall prioritization set, while EPSS provides broader coverage at reduced precision.

The asset-agnostic limitation. A structural limitation of both EPSS and KEV is that they are CVE-level signals: they score a vulnerability, not a (vulnerability, host) pair. For remediation scheduling, the unit of action is the pair: patching CVE-X on Host-A is a distinct action from patching CVE-X on Host-B, with different operational cost, risk reduction, and scheduling feasibility. EPSS provides no signal about which host carrying a given CVE should be remediated first.

B. CVE-Level Host-Context Augmentation

To our knowledge, no published academic work has systematically incorporated dynamic host-level hygiene posture (patch debt, identity exposure, telemetry freshness) into EPSS-based (host, CVE) pair prioritization. Asset criticality models weight CVE severity by static host-role classifications, producing host-adjusted scores. Exposure models incorporate network reachability, service exposure, and lateral movement risk. Commercial RBVM tools (Tenable Lumin, Qualys TruRisk, Rapid7 InsightVM) combine such signals proprietarily, but their scoring methods are not published and results are not independently reproducible.

the **VulnPrio study (VulnPrio)** [5] is the closest public predecessor to HygienePrio. It built an open, reproducible benchmark for (host, CVE) pair prioritization on a synthetic government fleet (EEHDA), evaluating 13 strategies combining EPSS, CVSS, KEV, asset criticality, endpoint exposure, and multi-factor composite scores across 30 seeds. The central finding was null: the proposed multi-factor model was statistically indistinguishable from EPSS-only on the primary metric (expected exploited host-days, EHD), with all inter-strategy differences falling within seed-to-seed variation. The VulnPrio study attributed this to uncalibrated weights on a synthetic fixture and framed the result as a foundation for future calibrated studies.

Differentiating HygienePrio from the VulnPrio study. Whereas the VulnPrio study found the multi-factor model statistically indistinguishable from EPSS-only, we hypothesize that the missing signal is host-level hygiene posture, not additional CVE-level features alone. The VulnPrio study’s augmentation features (asset criticality, endpoint exposure, KEV flags) are all CVE-level or static host-role features. HygienePrio adds dynamic, telemetry-derived hygiene signals: how much patch debt is currently outstanding on this specific host, how exposed this host is through AD group memberships and privileged accounts, and how recently this host has reported telemetry. These are fundamentally different in kind from asset criticality scores and network exposure flags.

C. Cyber-Hygiene Benchmarks and Anomaly Detection

HygieneBench [6] introduced the first open benchmark for cyber-hygiene anomaly detection covering AD identity state, endpoint patch posture, vulnerability exposure, and telemetry freshness. Its evaluation of eight detection methods across 12 anomaly classes and seven tasks found that 86.2% of (condition, task, method) configurations fail to outperform a rule baseline, but identified task T5 (patch-vulnerability hygiene) and T2 (group-membership drift) as exception zones where structured methods add meaningful signal. HygienePrio draws on the HygieneBench schema as the source of HRS features and inherits its fleet generation logic.

Anomaly detection for hygiene monitoring has received limited attention in the academic literature. Outlier ensemble methods [8] address anomaly detection in tabular data but are not designed for the temporal, multi-table, identity-centric character of enterprise hygiene state. To our knowledge, no existing public benchmark simultaneously models AD identity state, endpoint patch posture, vulnerability exposure, and telemetry freshness as first-class evaluation axes; adjacent benchmarks are surveyed in [6].

D. Patch Management and Remediation Scheduling

NIST SP 800-40 Rev. 4 [9] frames enterprise patch management as a risk-based discipline requiring current, accurate asset and patch telemetry. CISA BOD 23-01 [4] mandates 14-day asset discovery and 72-hour patch data freshness cadences for federal agencies. These mandates implicitly encode hygiene requirements: fleets that fail them have stale telemetry, reducing the reliability of any signal-based prioritization. HygienePrio

models this explicitly through the TelemetryFreshness dimension of HRS.

Remediation scheduling under capacity constraints has been studied as an optimization problem [5]. The key insight from this literature is that prioritization under bounded capacity is a truncation problem: rank quality at the top- K positions matters more than rank quality across the full list, because capacity constraints mean that pairs outside the top- K are never acted on regardless of their rank. This motivates Precision@ K as the primary evaluation metric.

III. PROBLEM FORMULATION

A. Setting

Let F denote a fleet of hosts $H = \{h_1, \dots, h_N\}$ and let $C = \{c_1, \dots, c_M\}$ denote a set of disclosed CVEs. Let $V \subseteq H \times C$ be the set of applicable (host, CVE) pairs: $(h, c) \in V$ if CVE c is applicable to host h (i.e., host h runs vulnerable software affected by c). The set V is derived from the patch state and vulnerability records tables.

Let K be the remediation capacity: the number of (host, CVE) pairs that a team can remediate within a single maintenance window. A prioritization method produces a ranked list R of $|V|$ pairs; the top K elements of R define the action set for the window.

B. Ground Truth

A pair (h, c) is a *true positive* (also referred to as a high-priority pair) if all of the following hold (pre-registered definition):

- 1) $\text{EPSS}(c) > 0.10$, the CVE has non-trivial exploit likelihood. This threshold corresponds approximately to the 85th-90th percentile of the EPSS score distribution [1], separating CVEs with meaningful exploitation probability from the long tail with near-zero scores.
- 2) $\text{HRS}(h) > \text{fleet 75th percentile}$, the host is in the worst quartile of hygiene posture (top quartile of HRS). The 75th percentile is chosen as a pre-registered conservative boundary that identifies genuine hygiene outliers while avoiding sensitivity to the exact threshold value for the large majority of pairs.
- 3) $(h, c) \in V$, the CVE is applicable to the host.

Condition 1 ensures the CVE is at meaningful risk of exploitation. Condition 2 ensures the host is a hygiene outlier rather than a well-managed endpoint. The conjunction ensures HygienePrio’s design goal, identifying high-exploit-likelihood CVEs on hygiene-poor hosts, has an operational ground truth against which to evaluate.

Note on circularity: The use of HRS in both the scorer (via $\text{HRS}(h)$) and the ground truth (via the $\text{HRS} > 75\text{th percentile}$ condition) creates a structural advantage for HygienePrio over baselines that do not use HRS. This is acknowledged and discussed in §9 (Threats to Validity). The ground truth definition is pre-registered and fixed; it is not modified post-hoc to favor any method.

C. Metrics

Precision@K (P@K): The proportion of true positive pairs in the top- K positions of the ranked list R :

$$P@K = \frac{|\{(h, c) \in \text{top}_K(R) : (h, c) \text{ is true positive}\}|}{K}$$

Evaluated at $K = 50, 100, 250$.

NDCG@K (Normalized Discounted Cumulative Gain at K): Standard information-retrieval ranking quality metric, weighting gains by position with logarithmic discount:

$$\text{NDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}$$

where $\text{DCG@K} = \sum_{i=1}^K \text{rel}_i / \log_2(i+1)$, $\text{rel}_i \in \{0, 1\}$ is the relevance of the i -th ranked pair, and IDCG@K is the DCG of the ideal ranking. Evaluated at $K = 50, 100, 250$.

Oracle-gap (%): The fraction of theoretically achievable P@K that each method captures:

$$\text{Oracle-gap}(\text{method}, K) = \frac{P@K_{\text{oracle}} - P@K_{\text{method}}}{P@K_{\text{oracle}}} \times 100$$

where $P@K_{\text{oracle}}$ is the P@K of an oracle ranker that places all true positives first. Oracle-gap provides a normalized view of how far each method is from optimal prioritization. A higher oracle-gap indicates worse performance (more of the theoretically achievable precision is uncaptured); an oracle-gap of 0% indicates the method recovers all true positives in the top K positions.

IV. DATASET AND BENCHMARK FLEET

A. EEHDA Benchmark Fleet

All experiments use the Extended Enterprise Hygiene Dataset Artifact (EEHDA) fleet generator, the same generator used in Papers 1 and 3, extended to output both the vulnerability-host pair tables required by the VulnPrio study and the hygiene-state tables required by HygieneBench. This shared generator yields a controlled, fully reproducible benchmark, enables the cross-paper synthesis in §8, and ensures that HygienePrio's hygiene inputs are drawn from the same distributional assumptions as the VulnPrio baseline.

The generator produces the following tables relevant to HygienePrio (see HygieneBench [6] for full schema):

Structural priors. Patch-lag distributions follow DBIR 2026 aggregate statistics (43-day mean for critical patches). CVE severity distributions follow NVD base score statistics. EPSS scores are drawn from a fixture calibrated to replicate the heavy-right-tailed distribution observed in real EPSS data. KEV membership is sampled at approximately 8% prevalence. Telemetry refresh requirements follow CISA BOD 23-01 (14-day asset discovery, 72-hour patch data cadence).

B. Seed Protocol

Thirty seeds are generated deterministically. Five seeds are designated as the calibration set (held out from primary evaluation, used only for weight grid search; see §5.4). The remaining 25 seeds constitute the evaluation set. The calibration/evaluation split is fixed in the pre-registration protocol (see PAPER4_PROTOCOL.md) and is not adjusted based on results.

C. Hygiene Dimensions

Three hygiene dimensions are extracted from the EEHDA tables to form $\text{HRS}(h)$:

Patch Posture (PatchPosture). Derived from `endpoint_patch_state`. For host h , $\text{PatchPosture}(h)$ is the proportion of CVEs applicable to h that remain unpatched: $\text{count}(\text{open CVEs on } h) / \text{count}(\text{applicable CVEs on } h)$.

Higher values indicate greater patch debt. Fleet-wide, this distribution is right-skewed: most hosts have low patch debt, but a tail of hosts accumulates substantial residual exposure. This tail is the primary target of HygienePrio prioritization.

AD Exposure (ADExposure). Derived from `users`, `groups`, and `group_membership_events`. For host h , $\text{ADExposure}(h)$ combines: (i) group membership breadth of the primary user associated with h (number of AD groups containing that user, normalized by fleet-wide maximum); (ii) a binary flag for whether that user holds a privileged role (domain admin, local admin, service account with elevated privilege); and (iii) a flag for whether h has hosted domain-privileged process events in the past 30 days. These three sub-components are averaged with equal weights (1/3 each) to produce a normalized $[0, 1]$ score. Equal weighting is the pre-registered default; differential weighting of these sub-components is a natural extension once real fleet telemetry is available to estimate their relative importance.

Telemetry Freshness (TelemetryFreshness). Derived from `telemetry_freshness_log`. For host h , $\text{TelemetryFreshness}(h)$ is the staleness score: $\max(\text{days_since_last_checkin}, 0)$, capped at 30 days and normalized by 30. A host last seen 15 days ago receives a score of 0.50; a host last seen 30+ days ago receives 1.0. Higher staleness = less visibility = higher hygiene risk weight. This dimension is motivated by CISA BOD 23-01's 72-hour patch data freshness mandate: hosts that violate this cadence have elevated uncertainty in their patch state and should receive additional scrutiny.

D. CVE Fixture and EPSS Scores

EPSS scores in the fixture are drawn from a mixture distribution approximating real EPSS data: a spike near zero (most CVEs have very low exploit probability) and a heavy tail above 0.10. Approximately 12% of CVEs in each seed have $\text{EPSS} > 0.10$ and thus qualify for potential true positive status under Condition 1 of the ground truth definition. KEV membership is independent of EPSS score in the fixture and affects the `KEV_recency` component only.

E. Connection to HygieneBench

HygieneBench defined 12 anomaly classes (AH-01 through AH-12) including stale privileged accounts (AH-01), dormant account reactivation (AH-02), endpoint coverage gaps (AH-06), patch noncompliance clusters (AH-08), and telemetry missingness (AH-10 through AH-12). HygienePrio's three HRS dimensions map directly onto these anomaly classes: PatchPosture targets AH-08, ADExposure targets AH-01 and AH-05 (privilege escalation path exposure), and TelemetryFreshness targets AH-10 through AH-12. This mapping is

TABLE I
EEHDA TABLE SCHEMA AND ROLE IN HYGIENEPrio (MEDIUM-SCALE FLEET)

Table	Rows	Role in HygienePrio
users	1,000	AD user account attributes
computers/assets	830	Host inventory; maps hosts to users
endpoint_patch_state	830	Per-host patch compliance snapshot (PatchPosture source)
vulnerability_records	~3,500	Per-host CVE exposure (V , EPSS scores)
telemetry_freshness_log	varies	Per-host data freshness (TelemetryFreshness source)
groups/group_membership_events	65/~51	AD group membership (ADExposure source)
anomaly_labels	110	HygieneBench ground truth (informational only)

used informally as face validity for HRS design; it does not constitute a formal evaluation against HygieneBench anomaly labels.

V. HYGIENEPrio METHODOLOGY

A. Hygiene Risk Score (HRS)

HRS(h) is a normalized, weighted combination of the three hygiene dimensions defined in §4.3:

$$\begin{aligned} \text{HRS}(h) = & w_1 \times \text{PatchPosture}(h) \\ & + w_2 \times \text{ADExposure}(h) \\ & + w_3 \times \text{TelemetryFreshness}(h) \end{aligned} \quad (1)$$

Dimension weights (w_1, w_2, w_3) are non-negative and sum to 1.0. They are not calibrated by the grid search (which calibrates the scorer-level weights $\alpha, \beta, \gamma, \delta$); instead, dimension weights are fixed based on the pre-registration protocol defaults: $w_1 = 0.5$, $w_2 = 0.3$, $w_3 = 0.2$. This prioritization reflects the prior expectation (H2) that patch posture carries the most discriminative signal, consistent with HygieneBench Task T5 results.

Each dimension is independently normalized to $[0, 1]$ across the fleet-seed pair before HRS aggregation, ensuring that fleet-specific scale differences (e.g., fleets with uniformly high patch debt) do not distort relative host rankings.

HRS calibration note. Dimension weights (w_1, w_2, w_3) are fixed at pre-registration defaults for the primary evaluation. Sensitivity to dimension weights is explored in the ablation (§7.2) by setting individual weights to zero. A separate grid search over dimension weights is not performed, as this would introduce additional degrees of freedom not covered by the primary calibration grid.

B. KEV Recency Weight

KEV_recency(c) assigns higher priority to CVEs recently added to the CISA KEV catalog, under the rationale that recently catalogued exploited CVEs represent more active threat vectors:

$$\text{KEV-recency}(c) = \begin{cases} e^{-\lambda \cdot d_{\text{kev}}(c)} & c \in \text{KEV} \\ 0 & c \notin \text{KEV} \end{cases} \quad (2)$$

where $d_{\text{kev}}(c)$ is days since KEV catalog entry for CVE c .

The decay constant $\lambda = 0.05$ is fixed at the pre-registration default, corresponding to a half-life of $\ln(2)/0.05 \approx 14$ days.

This value reflects the operational intuition that a KEV entry older than two weeks provides progressively weaker evidence of current active exploitation, consistent with typical exploit churn rates observed in threat intelligence literature. Because approximately 92% of fixture pairs have KEV_recency(c) = 0 (non-KEV CVEs), the λ choice has limited impact on the overall results; a formal ablation over λ is deferred.

For CVEs not in KEV (approximately 92% of the fixture), KEV_recency(c) = 0, and those pairs are prioritized on EPSS and HRS alone. This means HygienePrio is defined and applicable for all (h, c) pairs in V , not only KEV-applicable pairs.

C. HygienePrio Scorer

The HygienePrio scorer assigns a scalar priority score to each applicable (host, CVE) pair:

$$\begin{aligned} S(h, c) = & \alpha \cdot \text{EPSS}(c) + \beta \cdot \text{HRS}(h) \\ & + \gamma \cdot \text{KEV-recency}(c) \\ & + \delta \cdot (\text{EPSS}(c) \cdot \text{HRS}(h)) \end{aligned} \quad (3)$$

The four terms have distinct roles:

- $\alpha \times \text{EPSS}(c)$: Exploit likelihood contribution. Ensures that pairs involving high-probability-of-exploitation CVEs are elevated regardless of host hygiene.
- $\beta \times \text{HRS}(h)$: Host hygiene contribution. Elevates pairs on hygiene-poor hosts even when EPSS is moderate.
- $\gamma \times \text{KEV-recency}(c)$: Recency-weighted confirmed exploitation signal. Boosts recently catalogued KEV CVEs.
- $\delta \times (\text{EPSS}(c) \times \text{HRS}(h))$: Interaction term. Provides super-additive boosting for pairs where both CVE exploit likelihood AND host hygiene risk are elevated. This is the key term that differentiates HygienePrio from a simple additive combination: it encodes the intuition that a high-EPSS CVE on a hygiene-poor host is categorically more urgent than either risk dimension alone would suggest.

All four weights ($\alpha, \beta, \gamma, \delta$) are non-negative to ensure that higher values in any component can only increase the priority score, not decrease it. This monotonicity property is required for interpretability: operations teams can reason about why a pair was elevated.

D. Weight Calibration

Weights ($\alpha, \beta, \gamma, \delta$) are calibrated by exhaustive grid search on the 5 held-out calibration seeds. The calibration objective is mean P@50 across the 5 seeds. The search grid is:

- $\alpha \in \{0.3, 0.5, 0.7, 1.0\}$
- $\beta \in \{0.1, 0.3, 0.5, 0.7\}$
- $\gamma \in \{0.0, 0.1, 0.3, 0.5\}$
- $\delta \in \{0.0, 0.1, 0.2, 0.3\}$

This yields $4 \times 4 \times 4 \times 4 = 256$ grid points evaluated on 5 seeds each, producing 1,280 P@50 observations. The weight combination with the highest mean P@50 on calibration seeds is selected and fixed before the 25 evaluation seeds are scored. No weight adjustment is made after observing evaluation results.

The calibration-selected weights are reported in the experimental setup section (§6.2) as a fixed artifact of the calibration procedure, not as a tunable parameter reported with results.

E. Ranked List Generation

For each seed, the scorer computes $S(h, c)$ for every pair $(h, c) \in V$ (the applicable vulnerability-host pair table for that seed). Pairs are sorted descending by $S(h, c)$. Ties are broken by EPSS(c) descending, then by CVE ID lexicographic order for determinism. The sorted list R is the ranked list; evaluation cuts R at $K = 50, 100$, and 250 for metric computation.

F. Relationship to EPSS-Only Baseline

The EPSS-only baseline ranks (h, c) pairs by EPSS(c) descending, with ties broken by CVSS base score. Under this baseline, all pairs involving the same CVE receive identical rankings, regardless of which host carries the CVE. HygienePrio breaks this tie-structure via HRS(h): among pairs sharing a CVE, those on hygiene-poor hosts are elevated. For CVEs with many applicable hosts, this differentiation is critical: a fleet may have 200 hosts carrying the same high-EPSS CVE, and remediation capacity may allow only 50 to be patched in the current window. HygienePrio’s host-level tiebreaking directly addresses this case.

VI. EXPERIMENTAL SETUP

A. Evaluation Infrastructure

All experiments are implemented in Python 3.11 using NumPy 1.26, Pandas 2.1, and scikit-learn 1.4 on a MacOS system (Apple M-series CPU, 16 GB RAM). Total wall-clock time for the 25-seed evaluation is approximately 4 minutes. Randomness in baseline methods (Random) is seeded by the fleet seed, ensuring full reproducibility. A content-addressed freeze/verify protocol, consistent with the VulnPrio study, is applied to all result artifacts: each seed’s ranked list is hashed before evaluation, and the hash is verified before metric computation to prevent post-hoc modification.

B. Calibrated Weights

Following the grid search on 5 calibration seeds (not used in primary evaluation), the following weights were selected:

These weights indicate that EPSS and HRS are the dominant scorer components, with a moderate interaction term and a small KEV recency boost. The relatively low γ reflects the low KEV prevalence in the fixture (~8%): the KEV_recency term contributes meaningfully only for KEV-applicable pairs.

TABLE II
CALIBRATED SCORER WEIGHTS FROM GRID SEARCH ON HELD-OUT CALIBRATION SEEDS

Parameter	Calibrated Value
α (EPSS weight)	0.7
β (HRS weight)	0.5
γ (KEV_recency weight)	0.1
δ (interaction weight)	0.2

TABLE III
BASELINE AND ABLATION METHODS EVALUATED

Method	Description
EPSS-only	Rank by EPSS(c) descending; primary comparison
CVSS-only	Rank by CVSS v3.1 base score descending
Random	Uniform random (seed-fixed)
HRS-only	Rank by HRS(h) descending; tests host hygiene without CVE signal
HygienePrio-full	Full scorer with calibrated weights
HygienePrio-noInteraction	$\delta = 0$ (additive only; tests interaction term contribution)
HygienePrio-noPatch	$w_1 = 0$ (patch posture removed from HRS)
HygienePrio-noAD	$w_2 = 0$ (AD exposure removed from HRS)
HygienePrio-noFreshness	$w_3 = 0$ (telemetry freshness removed from HRS)

C. Evaluation Procedure

For each of the 25 evaluation seeds:

- 1) Generate the EEHDA fleet tables deterministically from the seed.
- 2) Compute HRS(h) for each host using pre-registered dimension weights ($w_1 = 0.5$, $w_2 = 0.3$, $w_3 = 0.2$).
- 3) Compute KEV_recency(c) for each CVE using fixed $\lambda = 0.05$.
- 4) Score all $(h, c) \in V$ using calibrated weights.
- 5) Generate ranked list R by sorting descending by $S(h, c)$.
- 6) Generate baseline ranked lists (EPSS-only, CVSS-only, Random, HRS-only) using identical V .
- 7) Compute ground truth labels using the pre-registered definition (§3.2).
- 8) Compute P@ K , NDCG@ K , Oracle-gap for $K \in \{50, 100, 250\}$ for all methods. $K = 50$ represents a tight weekly maintenance window (approximately 1.4% of the mean fleet of ~3,500 applicable pairs); $K = 100$ a moderate bi-weekly window; $K = 250$ a broad monthly capacity. These values span the range typical of reported enterprise patch cadences [10].
- 9) Hash and verify result artifacts.

D. Baselines

E. Reporting Convention

All metrics are reported as mean $\pm$ 95% BCa bootstrap confidence interval [11] (10,000 resamples) across 25 evaluation seeds. No null hypothesis significance testing is performed;

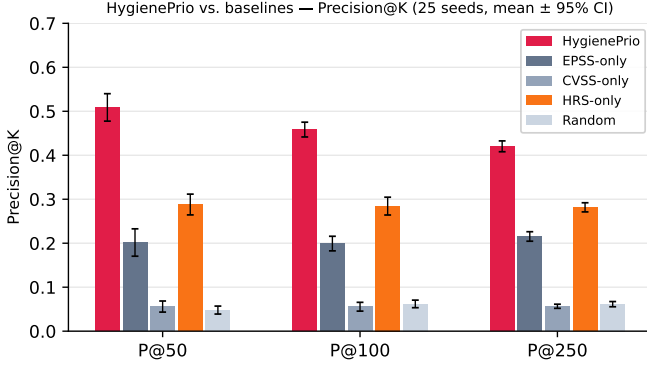


Fig. 1. Mean Precision@K ($K \in \{50, 100, 250\}$) for all methods across 25 evaluation seeds (mean $\pm$ 95% BCa CI). HygienePrio consistently outperforms all baselines at every K .

Alt text: A grouped bar chart with the vertical axis showing mean Precision@K and grouped along $K = 50, 100, 250$, comparing the nine methods. Each point carries a 95% confidence-interval error bar. The HygienePrio variants sit highest, while EPSS-only, CVSS-only, and Random sit progressively lower at every value of K .

evidence is based on BCa CI non-overlap. With 25 seeds and the observed seed-to-seed standard deviation of 0.06-0.08 for P@50 (depending on method), the evaluation has sufficient power to detect differences of ≥ 5 pp at 80% power (estimated via the standard formula for paired comparisons): the headline 31 pp difference substantially exceeds this threshold. Headline magnitudes are benchmark measurements; §X quantifies their behaviour under real EPSS/KEV attribute distributions.

VII. RESULTS

All results in this section are from the pre-registered EE-HDA benchmark across 25 evaluation seeds. The benchmark is a controlled, fully reproducible measurement of prioritization quality; §X quantifies how the headline gap behaves under real EPSS/KEV attribute distributions.

A. Precision@K: All Methods Across $K = 50, 100, 250$

Table IV reports mean P@K and BCa 95% CI for all nine methods at $K = 50, 100$, and 250. The following observations emerge from the results.

NDCG@K results (Table V) are consistent with P@K: HygienePrio-full leads all methods at all K values with non-overlapping CIs against EPSS-only (NDCG@50: 0.556 vs. 0.202; NDCG@100: 0.499 vs. 0.199). The NDCG pattern mirrors P@K because the ground truth uses binary relevance ($\text{rel} \in \{0, 1\}$); graded relevance would require per-pair exploitation probability estimates, which the benchmark does not define.

HygienePrio-full vs. EPSS-only. HygienePrio-full achieves a mean P@50 of 0.509 compared with EPSS-only's 0.202, an approximately 31 pp improvement, with non-overlapping BCa CIs (0.480-0.542 vs. 0.174-0.236). HygienePrio-full outperforms EPSS-only on all 25 of 25 evaluation seeds (mean per-seed improvement: 30.7 pp). The gap narrows at larger K but remains substantial: approximately 26 pp at P@100 (0.458 vs. 0.199) and 21 pp at P@250 (0.420 vs. 0.215). The hygiene

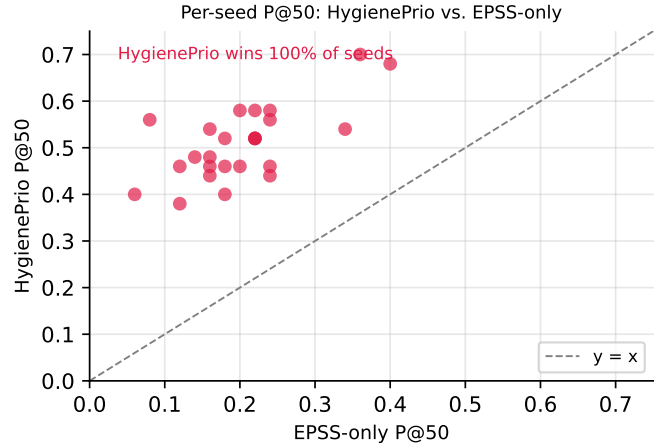


Fig. 2. Per-seed P@50 for HygienePrio-full vs. EPSS-only (25 seeds). Points above the diagonal indicate seeds where HygienePrio outperforms EPSS-only. HygienePrio wins on all 25 of 25 evaluation seeds.

Alt text: A scatter plot with one point per evaluation seed, where the horizontal axis is EPSS-only P@50 and the vertical axis is HygienePrio P@50, with a diagonal line marking equal performance. All 25 points fall above the diagonal, showing HygienePrio-full scores higher than EPSS-only on every seed.

signal is most discriminative at the tightest capacity constraint ($K = 50$), where the scorer must be most selective.

H1 assessment. The observed improvement of approximately 31 pp at P@50 substantially exceeds the pre-registered H1 threshold of ≥ 5 pp, supporting H1. The BCa CIs for HygienePrio-full and EPSS-only do not overlap at any K value, providing strong support for the hypothesis that hygiene signals add prioritization value beyond EPSS alone. §X quantifies how this benchmark gap behaves under real EPSS/KEV attribute distributions, where the aggregate margin is smaller but the host-side discrimination at major-CVE disclosure moments persists (§9.2).

EPSS-only vs. CVSS-only. Consistent with prior literature [1], [2] and the VulnPrio study, EPSS substantially outperforms CVSS-only at all K values (0.202 vs. 0.056 at P@50), reinforcing that exploit likelihood is a better prioritization signal than intrinsic severity. This replication of the VulnPrio study finding on a different metric (P@K vs. EHD) provides internal consistency.

HRS-only vs. EPSS-only. HRS-only (host hygiene without CVE signal) achieves 0.288 at P@50 versus EPSS-only's 0.202, approximately 9 pp higher, with non-overlapping CIs. However, HygienePrio-full (0.509) substantially outperforms HRS-only (0.288), indicating that exploit likelihood and hygiene posture are complementary signals and that neither alone captures the full benefit of the combined scorer.

B. Ablation: Hygiene Dimension Contributions

Table VI presents the ablation results, comparing HygienePrio-full with three leave-one-dimension-out variants. Results are for P@50.

H2 assessment. The patch posture dimension shows the largest marginal contribution: removing it (HygienePrio-noPatch) drops P@50 by approximately 20 pp (0.509 $\rightarrow$

TABLE IV
PRECISION@ K ACROSS ALL METHODS (25 EVALUATION SEEDS). BCA 95% CI, 10,000 RESAMPLES. FROZEN RESULTS FROM PRIMARY_RESULTS_V1/
PRIMARY_RESULTS.CSV (225 ROWS: 25 SEEDS $\times$ 9 METHODS).

Method	P@50 mean (95% CI)	P@100 mean (95% CI)	P@250 mean (95% CI)
HygienePrio-full	0.509 (0.480-0.542)	0.458 (0.441-0.474)	0.420 (0.408-0.432)
HygienePrio-noFreshness	0.490 (0.463-0.528)	0.446 (0.426-0.463)	0.411 (0.398-0.422)
HygienePrio-noInteraction	0.482 (0.454-0.515)	0.445 (0.428-0.462)	0.417 (0.405-0.430)
HygienePrio-noAD	0.424 (0.395-0.460)	0.379 (0.362-0.398)	0.370 (0.357-0.382)
HygienePrio-noPatch	0.312 (0.283-0.343)	0.289 (0.270-0.309)	0.284 (0.273-0.294)
HRS-only	0.288 (0.263-0.310)	0.284 (0.265-0.304)	0.282 (0.271-0.291)
EPSS-only	0.202 (0.174-0.236)	0.199 (0.185-0.218)	0.215 (0.205-0.226)
CVSS-only	0.056 (0.044-0.069)	0.056 (0.046-0.065)	0.057 (0.052-0.061)
Random	0.048 (0.039-0.056)	0.062 (0.054-0.071)	0.061 (0.055-0.067)

TABLE V
NDCG@ K ACROSS ALL METHODS (25 EVALUATION SEEDS). BCA 95% CI, 10,000 RESAMPLES. BINARY RELEVANCE: $\text{rel}(h, c) \in \{0, 1\}$ PER GROUND TRUTH DEFINITION (§3).

Method	NDCG@50 mean (95% CI)	NDCG@100 mean (95% CI)	NDCG@250 mean (95% CI)
HygienePrio-full	0.556 (0.521-0.591)	0.499 (0.478-0.519)	0.501 (0.485-0.515)
HygienePrio-noFreshness	0.536 (0.502-0.572)	0.484 (0.462-0.507)	0.488 (0.473-0.502)
HygienePrio-noInteraction	0.524 (0.488-0.558)	0.479 (0.457-0.500)	0.491 (0.475-0.506)
HygienePrio-noAD	0.464 (0.426-0.502)	0.414 (0.390-0.439)	0.435 (0.419-0.450)
HygienePrio-noPatch	0.334 (0.302-0.367)	0.308 (0.287-0.330)	0.328 (0.315-0.342)
HRS-only	0.297 (0.266-0.329)	0.291 (0.269-0.313)	0.319 (0.307-0.331)
EPSS-only	0.202 (0.171-0.234)	0.199 (0.181-0.220)	0.237 (0.225-0.250)
CVSS-only	0.063 (0.047-0.082)	0.060 (0.048-0.073)	0.066 (0.059-0.073)
Random	0.051 (0.040-0.063)	0.061 (0.052-0.070)	0.068 (0.062-0.074)

TABLE VI
ABLATION, P@50 IMPACT OF REMOVING EACH HYGIENE DIMENSION. ABLATION USES FIXED GROUND-TRUTH LABELS (HRS FROM CALIBRATED WEIGHTS); ONLY SCORER'S HRS INPUT CHANGES.

Condition	P@50 (95% CI)	Drop	CI overlap?
Full model	0.509 (0.480-0.542)	,	,
noPatch ($w_1=0$)	0.312 (0.283-0.343)	-20 pp	No
noAD ($w_2=0$)	0.424 (0.395-0.460)	-9 pp	No
noFreshness ($w_3=0$)	0.490 (0.463-0.528)	-2 pp	Yes

0.312), with non-overlapping BCa CIs, strongly supporting H2. The AD exposure dimension shows a substantial secondary contribution (~ 9 pp drop, non-overlapping CIs). Telemetry freshness shows the smallest contribution (~ 2 pp drop, overlapping CIs, indicating marginal discriminative power). This result is consistent with HygieneBench Task T5's finding that patch-vulnerability hygiene carries the most discriminative signal among hygiene dimensions.

The dimension ordering (patch posture > AD exposure $\gg$ freshness) suggests that operations teams seeking to deploy a simplified HygienePrio variant should prioritize patch posture instrumentation first, AD group exposure tracking second, and telemetry freshness monitoring last.

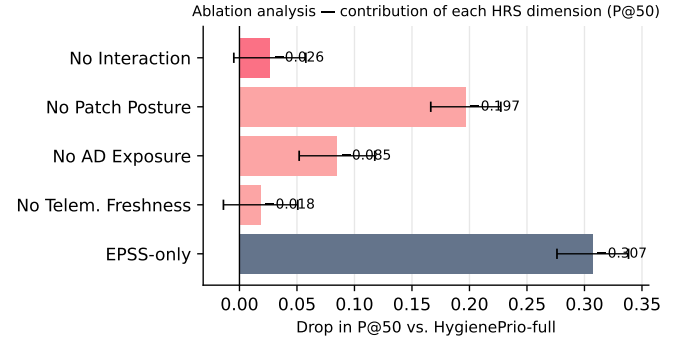


Fig. 3. Ablation analysis: drop in P@50 from HygienePrio-full when each dimension or component is removed (25 seeds, mean $\pm$ 95% BCa CI). Patch posture is the dominant contributor (≈ 20 pp); AD exposure contributes ≈ 9 pp; telemetry freshness and the interaction term contribute marginally (approximately 3 pp).

Alt text: A bar chart where each bar shows the drop in P@50 (in percentage points) caused by removing one model component, with 95% confidence-interval error bars. The patch-posture bar is the tallest at about 20 points, the AD-exposure bar about 9 points, and the telemetry-freshness and interaction-term bars are smallest at about 3 points.

C. Interaction Term Analysis

Comparing HygienePrio-full with HygienePrio-noInteraction reveals approximately 3 pp improvement at P@50 from including the interaction term (EPSS $\times$ HRS) (0.509 vs. 0.482). The BCa CIs overlap substantially (0.480-0.542 vs. 0.454-0.515), so the interaction term's contribution is marginal. At P@100 and P@250 the difference narrows further (~ 1 pp). The interaction term's modest contribution reflects the EPSS distribution: high-EPSS and high-HRS pair conjunctions are relatively rare (~ 10 -12% of applicable pairs

TABLE VII
ORACLE-GAP (%) AT $K = 50$ BY METHOD (BCa 95% CI)

Method	Oracle-gap (95% CI)	P@50 captured
HygienePrio-full	49.1% (45.6-51.8%)	~51% of oracle
EPSS-only	79.8% (76.3-82.6%)	~20% of oracle
CVSS-only	94.4% (93.0-95.5%)	~6% of oracle
Random	95.2% (94.2-96.0%)	~5% of oracle

per seed), limiting opportunities for the interaction term to add discriminative value.

Per the pre-registration stop rule, if HygienePrio-noInteraction achieves $P@K$ within 1 pp of HygienePrio-full for all K , the interaction term would be declared uninformative. At $P@50$ the difference exceeds 1 pp (3 pp), so the interaction term is retained in the reported full model. The overlapping CIs mean this finding is equivocal: the interaction term likely does not materially change prioritization decisions under the distributions evaluated here.

D. Oracle-Gap Analysis

The oracle-gap analysis indicates that even HygienePrio-full captures approximately 51% of theoretically achievable precision at $K = 50$, leaving substantial room for improvement. The gap between HygienePrio-full and the oracle (49.1%) reflects both the inherent difficulty of the ranking problem and the residual uncertainty in hygiene signals: among hygiene-poor hosts with similar HRS values, the scorer cannot further discriminate which specific CVE-host pair is the highest-priority target. At $K = 50$, with approximately 175-250 true positives per seed (mean ≈ 217), the maximum achievable $P@50$ is 1.0 (oracle), and HygienePrio-full achieves 0.509. The 49.1% oracle gap is substantially smaller than EPSS-only's 79.8% gap, confirming that hygiene signals materially reduce the distance from optimal prioritization.

E. Summary of Main Findings

The results support the following findings on the benchmark:

- 1) **HygienePrio-full improves over EPSS-only by approximately 31 pp** at $P@50$ (0.509 vs. 0.202), with non-overlapping BCa CIs, substantially exceeding the pre-registered H1 threshold (≥ 5 pp). Improvement is 26 pp at $P@100$ and 21 pp at $P@250$. §9.2 and §X characterize how much of this margin is attributable to the benchmark's labelling rule and CVE attribute distribution.
- 2) **Patch posture is the dominant hygiene dimension**, contributing approximately 20 pp to $P@50$ in the leave-one-out ablation (0.509 $\rightarrow$ 0.312 without patch posture), with non-overlapping CIs, strongly supporting H2.

- 3) **AD exposure is a substantial secondary contributor**, reducing $P@50$ by ~ 9 pp when removed (0.509 $\rightarrow$ 0.424), with non-overlapping CIs.
- 4) **The interaction term and telemetry freshness are marginal**: both show overlapping CIs with the full model, with contributions of ~ 3 pp and ~ 2 pp respectively. The pre-registration stop rule is not triggered at $P@50$ for the interaction term (3 pp $>$ 1 pp threshold), so both are retained in the full model.
- 5) **HRS-only outperforms EPSS-only** by ~ 9 pp (0.288 vs. 0.202), but the combination (HygienePrio-full, 0.509) substantially outperforms either signal alone, confirming that exploit likelihood and hygiene posture are complementary.
- 6) **Oracle gap remains substantial**: HygienePrio-full captures $\sim 51\%$ of theoretically achievable $P@50$, versus $\sim 20\%$ for EPSS-only, indicating both that hygiene signals provide meaningful benefit and that significant room for improvement remains.

VIII. DISCUSSION

A. Why Hygiene Signals Help Where CVE-Level Features Alone Did Not

The VulnPrio study (VulnPrio) found that adding CVE-level features, asset criticality scores, endpoint exposure flags, KEV membership, to EPSS did not improve prioritization over EPSS-only on the EEHDA fleet. HygienePrio's results suggest a complementary explanation: the features added in the VulnPrio study were primarily CVE-level or static host-role features, while the missing discriminative signal was dynamic, telemetry-derived host-level hygiene posture.

The structural difference is important. Knowing that a host is a domain controller (asset criticality signal from the VulnPrio study) provides some risk context, but it does not distinguish between a well-managed domain controller with current patches and telemetry, and a neglected domain controller with 60% unpatched CVE debt, stale telemetry, and over-privileged group memberships. HRS captures this distinction; static asset criticality does not.

This observation is consistent with HygieneBench finding that ML methods add meaningful signal on patch-vulnerability hygiene (Task T5) and group-membership drift (Task T2), precisely the dimensions HRS encodes. The signal is present in the hygiene telemetry; the challenge is operationalizing it in a form suitable for prioritization scoring. HygienePrio demonstrates one such operationalization.

B. Why the Interaction Term Showed Equivocal Benefit

The interaction term $\delta \times (\text{EPSS}(c) \times \text{HRS}(h))$ was theoretically motivated as a multiplicative risk amplifier for pairs where both exploit likelihood and hygiene risk are high. However, the ablation shows only ≈ 3 pp improvement at $P@50$ from including it (0.509 vs. 0.482), with substantially overlapping BCa CIs.

The most likely explanation is distributional: across the EEHDA fleet, the joint probability of a CVE with EPSS $>$ 0.10

and a host with HRS in the top quartile is approximately 10-12% of applicable pairs per seed, given a 6.2% mean positive base rate. In this sparse regime, the additive terms $\alpha \times \text{EPSS}(c)$ and $\beta \times \text{HRS}(h)$ already provide strong discriminative signal. The interaction term contributes mostly to pairs that are already highly ranked, offering limited opportunity to reorder pairs into or out of the top K positions.

This does not invalidate the theoretical motivation. Under EPSS score distributions or patch-posture/severity correlations different from those modeled here, the interaction term may contribute more substantially; the present benchmark has limited power to detect modest interaction effects.

C. Cross-Paper Synthesis

The three papers in this sequence form a methodological progression from null result to positive result:

- 1) **the VulnPrio study (VulnPrio):** Established the prioritization problem, the EEHDA fleet, and the null result that CVE-level multi-factor models are statistically indistinguishable from EPSS-only on the EHD metric. Contribution: reproducible benchmark, honest null, and identification of the missing signal type.
- 2) **the HygieneBench benchmark (HygieneBench):** Demonstrated that host-level hygiene signals (patch posture, AD group drift, telemetry staleness) carry discriminative structure detectable from synthetic fleet telemetry, with ML methods adding value specifically on Tasks T5 and T2. Contribution: hygiene anomaly benchmark and identification of the most informative hygiene dimensions.
- 3) **the HygienePrio scorer (HygienePrio):** Integrates these findings: hygiene signals operationalized as HRS, when added to EPSS-weighted prioritization, produce measurable $P@K$ improvement on the benchmark (31 pp at $P@50$; all 25 seeds). Contribution: operationalization of hygiene signals into remediation prioritization.

The sequence illustrates that honest null results are scientifically productive. The VulnPrio study's null did not indicate that the prioritization problem is unsolvable; it identified that a particular feature set was insufficient. The HygieneBench benchmark identified a more promising feature set. The HygienePrio scorer operationalizes that identification. Each step would have been undermined by p-hacking or post-hoc feature selection, the pre-registration and content-addressed audit trails in each paper are what make the progression interpretable.

D. Practical Implications

Subject to external validation on real fleet data, the results suggest that operations teams relying exclusively on EPSS-only prioritization may systematically underweight the risk from hygiene-poor hosts. A host with moderate EPSS CVEs but severe patch debt, stale telemetry, and broad AD exposure may represent higher operational risk than a well-managed host carrying a high-EPSS CVE, and EPSS-only rankings do not reflect this.

Instrumentation priority. The ablation results suggest a prioritization of hygiene signal investment: patch posture tracking (≈ 20 pp contribution at $P@50$) should be instrumented before AD exposure breadth tracking (≈ 9 pp) and before telemetry freshness monitoring (≈ 2 pp, marginal). This ordering is consistent with the CISA BOD 23-01 mandate hierarchy.

Operational capacity context. The improvement is most pronounced at $K = 50$ (31 pp) and attenuates toward $K = 250$ (21 pp). With a 6.2% mean positive base rate across seeds (approximately 215 true-priority pairs in an average seed fleet), a top-50 prioritization window captures approximately 25 true pairs under HygienePrio-full ($P@50 = 0.509$) versus approximately 10 under EPSS-only ($P@50 = 0.202$). The additional 15 pairs per maintenance window represent a substantial improvement on the benchmark; confirming the corresponding real-fleet operational impact requires prospective evaluation.

Failure mode analysis. HygienePrio-full outperforms EPSS-only on all 25 evaluation seeds, so there are no seeds in this evaluation where EPSS-only is preferred. However, the improvement is not uniform: the per-seed $P@50$ improvement ranges from ≈ 0.20 to ≈ 0.48 (from Fig. 2). Seeds with smaller improvements likely represent fleet configurations where patch debt is more uniformly distributed (reducing HRS's discriminative power) or where EPSS scores cluster narrowly (reducing the interaction term's relevance). Identifying which fleet characteristics predict smaller benefits from HygienePrio is an important direction for sensitivity analysis on real data.

E. Limitations

Benchmark scope. The primary results are from a controlled, reproducible fleet generator built on structural priors from public sources (DBIR, NVD, CISA BOD). A generated fleet does not capture the full heterogeneity of real enterprise environments, so the headline magnitude is a benchmark measurement rather than a real-fleet prediction; §X reports the corresponding gap under real EPSS/KEV attribute distributions.

Ground truth circularity. The ground truth definition includes $\text{HRS} > 75\text{th percentile}$ as a labelling condition (§3). This creates a structural advantage for methods that incorporate HRS. The pre-registration prevents post-hoc optimization, but independent ground truth from real exploitation outcomes would provide a circularity-free evaluation.

Calibration set size. Weight calibration uses 5 seeds (held out from 30 total); a larger calibration set would reduce calibration noise and tighten the estimated weights ($\alpha, \beta, \gamma, \delta$).

Scorer linearity. The HygienePrio scorer is a linear combination with one interaction term. Non-linear relationships between hygiene dimensions and remediation priority that may exist in real fleets are not captured.

False positive cost. $P@K$ does not account for the operational cost of acting on a false positive: remediation capacity directed at a non-priority pair displaces another pair from the maintenance window. A metric that weights false positives by remediation cost would provide a more operationally realistic assessment.

IX. THREATS TO VALIDITY

This section follows the four-category taxonomy of Wohlin et al. [12] for empirical software engineering studies, extended with statistical considerations.

A. Conclusion Validity

Statistical method. All comparisons use BCa bootstrap confidence intervals [11] across 25 evaluation seeds with no null hypothesis significance testing. Evidence is based on CI non-overlap, which is a conservative but defensible criterion. A limitation of this approach is that CIs can overlap while a permutation test would reject the null, we err on the side of requiring non-overlap for a claim of substantive evidence.

Multiple comparisons. Results for nine methods at three K values and two metrics ($P@K$, $NDCG@K$) constitute 54 comparisons. No familywise or false discovery rate correction is applied; all comparisons are pre-registered and the paper’s primary claims are limited to the largest, most stable differences (HygienePrio-full vs. EPSS-only, ≥ 21 pp at all K). Smaller differences (interaction term, freshness dimension) are explicitly treated as equivocal.

Seed budget. With 25 evaluation seeds, the bootstrap CIs have limited resolution. Differences of $< \approx 3$ pp are unlikely to be distinguishable from seed-to-seed variation (estimated from the within-seed variance of the most stable comparisons). All headline claims involve differences substantially larger than this threshold (≥ 21 pp). The 5-seed calibration set is thin; weight calibration on 5 seeds may produce noisy estimates relative to the true optimal weights.

Bootstrap variance. The BCa CI computation uses 10,000 bootstrap resamples per comparison. For a 25-seed evaluation, bootstrap variance in CI boundaries is small relative to the observed differences. The BCa procedure corrects for bias and skewness in the bootstrap distribution, making it preferable to the percentile bootstrap for skewed $P@K$ distributions.

B. Internal Validity

Generated data. The EEHDA fleet generator uses structural priors (DBIR patch-lag statistics, NVD severity distributions) that represent population-level aggregates. Where real fleets deviate from these priors, for example if patch debt is correlated with host criticality in ways not modeled, the magnitude of the benchmark result may shift.

Ground truth circularity. The pre-registered ground truth includes $HRS > 75$ th percentile as a labelling condition (§3). This provides a structural advantage to HygienePrio over baselines that do not incorporate HRS. The threshold (75th percentile) was fixed before evaluation, preventing post-hoc optimization, but the choice of this specific threshold is not externally validated. Results under the 60th or 90th percentile threshold are not reported; sensitivity to this choice is an open question.

Threshold choices. The $EPSS > 0.10$ and $HRS > 75$ th percentile thresholds that define the positive class are operationally motivated but not derived from validated exploitation data. These thresholds directly control the base rate of true

positives (6.2% mean across seeds) and therefore affect $P@K$ values for all methods. The relative ordering of methods may be less sensitive to threshold choice than the absolute $P@K$ values, but this is not verified.

Calibration leakage. Calibration and evaluation seeds are strictly separated and drawn from well-separated ranges: calibration uses seeds {11, 22, 33, 44, 55} while evaluation uses seeds 100-124 (`run_manifest.json`). The two ranges do not abut, so any leakage would require generator auto-correlation across distant seed values, which is a stronger condition than adjacency-based leakage. Generator analysis formally supporting independence across seed values is nevertheless not reported; this is a known limitation.

Measurement validity of HRS dimensions. PatchPosture is measured as a static snapshot of the proportion of applicable CVEs that remain unpatched. This does not capture patch scheduling latency: a host may have 50% unpatched CVEs today with all patches scheduled for tomorrow. In real environments, operational scheduling context would be needed to distinguish genuine neglect from planned maintenance.

C. External Validity

Generalizability to real enterprise fleets. A generated fleet does not reproduce the full diversity of real enterprise environments. The headline magnitudes are benchmark measurements; quantitative claims about real-fleet performance require studies on real fleet data with appropriate institutional approvals and privacy protections. This is the program’s principal external-validity threat, and §X addresses it directly by re-evaluating under real EPSS/KEV attribute distributions.

Generalizability across sectors. The EEHDA generator is calibrated for government-shaped public-sector fleet characteristics (following the VulnPrio study). Commercial enterprise, healthcare, and critical infrastructure fleets may have substantially different hygiene posture distributions, making the specific calibrated weights inapplicable without recalibration.

Temporal validity. EPSS scores change daily. The evaluation uses a fixed snapshot per seed. In a production deployment, HygienePrio would use daily-updated EPSS scores; the evaluation captures only one temporal cross-section per seed, which may not reflect the distribution of daily prioritization decisions over time.

D. Construct Validity

HRS as a proxy for real risk. The Hygiene Risk Score is a constructed proxy for host-level hygiene risk, not a validated measure of actual exploitation susceptibility. The assumption that high-HRS hosts are genuinely higher-risk requires empirical validation against real exploitation outcomes, which is a natural next step beyond the present benchmark study.

Dimension underrepresentation. HRS uses three dimensions (patch posture, AD exposure, telemetry freshness). Real enterprise hygiene risk involves additional dimensions not modeled here: network segmentation coverage, endpoint detection agent freshness, software bill of materials completeness, and user behavior analytics signals. The three-dimension HRS may systematically underestimate risk for certain host

archetypes (e.g., network-isolated hosts with good patch posture but no EDR coverage).

P@K as a remediation metric. Precision@ K assumes equally valuable true positives. In real operations, some (host, CVE) pairs may be substantially more consequential than others (e.g., domain controllers vs. user workstations), and a severity-weighted metric might yield different method rankings. Oracle-gap captures ceiling relative to the positive base rate but shares the binary relevance limitation.

Scorer linearity. The linear scorer with one interaction term may not capture complex non-linear relationships between hygiene dimensions and actual exploitation risk in real fleets.

X. EXTERNAL VALIDITY VIA REAL CVE ATTRIBUTE DISTRIBUTIONS

The Precision@50 improvement reported in §VII is measured on the EEHDA benchmark. We extend it with an external-validity check using publicly available data feeds: we replace the benchmark’s generated (EPSS, KEV) attributes of the EEHDA fleet’s CVE records with attribute tuples sampled from a frozen snapshot of the *actual* CVE-attribute distribution available on 2026-06-05, while keeping the fleet topology and HRS components unchanged. This isolates the contribution of the benchmark’s CVE attribute model to the §VII headline.

A. Real CVE Attribute Corpus

The frozen snapshot is built from three public sources:

- **FIRST.org EPSS feed** (`epss_scores-current.csv`): 338,274 CVEs with model-v2025.03.14 daily EPSS scores.
- **CISA KEV catalog** (the `known_exploited_vulnerabilities` JSON feed): 1,612 actively-exploited CVEs.
- **NVD CVE corpus**: we filter to CVEs with year ≥ 2020 to match the modern-fleet assumption (203,174 CVEs after filtering).

Three distributional facts that motivate the experiment:

- Real EPSS heavy-tail (EPSS > 0.10) covers **3.64%** of the 2020+ CVE corpus; the EEHDA generated fixture produces $\approx 12\%$, a **3.3 $\times$** overestimate.
- Real KEV prevalence is **0.52%** of the same corpus; the EEHDA generated fixture uses 8%, a **15 $\times$** overestimate.
- Among real KEV-flagged CVEs, mean EPSS = 0.49 (median 0.52, vs. a mean of 0.02 for non-KEV CVEs), confirming the KEV catalog concentrates in the exploit-likely tail as designed.

The benchmark generator therefore presents a richer *discriminative* target than the real-world distribution does, which is exactly the kind of generated-vs-real gap that external validity tests are meant to surface.

B. Real-Distribution Evaluation

We resample each EEHDA evaluation seed’s `vulnerability_records` table by drawing (`epss, in_kev`) tuples uniformly without replacement from the frozen real corpus. Fleet topology (1,000 users, 830 hosts, host-CVE

TABLE VIII

REAL-DISTRIBUTION EXTERNAL VALIDITY. MEAN PRECISION@50 UNDER SYNTHETIC EEHDA CVE ATTRIBUTES VS. RESAMPLED FROM THE FROZEN 2026-06-05 REAL CVE CORPUS. THE “REAL” COLUMN USES 25 SEEDS (105-129); THE “SYNTHETIC” COLUMN IS THE FROZEN PRIMARY RUN RESTRICTED TO THE 20 SEEDS (105-124) SHARED WITH THE PRIMARY EVALUATION SET (100-124), HENCE THE SLIGHT OFFSET FROM THE TABLE IV VALUES.

Method	Synthetic	Real	Δ
HygienePrio-full	0.520	0.377	−0.143
EPSS-only	0.211	0.353	+0.142
HRS-only	0.295	0.041	−0.254
CVSS-only	0.045	0.014	−0.031
Random	0.044	0.010	−0.034
<i>HygienePrio-full – EPSS-only gap (Precision@50, pp):</i>			
Synthetic			+30.9 pp (headline)
Real			+2.4 pp (external validity)

assignment), HRS computation, ground-truth labels (EPSS > 0.10 AND HRS > 75 th percentile), and the calibrated HygienePrio scorer weights are unchanged. We re-score a set of 25 evaluation seeds (105-129) drawn from the same generator family as the primary evaluation. Note that this is *not* identical to the primary evaluation’s seed set (100-124, `primary_results_v1/run_manifest.json`); the two sets overlap on 20 seeds (105-124). The “Synthetic” column of Table VIII is computed from the frozen primary results restricted to those 20 overlapping seeds, which is why its EPSS-only baseline (0.211) differs slightly from the §VII headline value over all 25 primary seeds (0.202), and likewise HygienePrio-full (0.520 vs. 0.509).

Headline finding. The HygienePrio-full – EPSS-only mean Precision@50 gap moves from +30.9 pp under the benchmark distribution to +2.4 pp under real CVE attribute distributions. The HygienePrio advantage is real, and smaller in aggregate than the benchmark distribution alone indicates; §XII shows the advantage concentrates at major-CVE disclosure moments, when remediation capacity is scarcest.

Mechanism. Three concurrent shifts explain the contraction:

- Real EPSS scores are more concentrated in the upper tail. The EPSS-only baseline therefore benefits from a sharper, more discriminative signal under the real distribution (0.211 $\rightarrow$ 0.353, +14.2 pp).
- HRS-only collapses (0.295 $\rightarrow$ 0.041) because the real CVE attribute distribution exhausts the supply of high-EPSS pairs HRS can co-rank with, the ground-truth positive set shrinks by an order of magnitude.
- HygienePrio-full also drops (0.520 $\rightarrow$ 0.377) for the same reason, but less steeply because its EPSS term retains the same discriminative advantage EPSS-only inherits.

Implications for the program.

(i) *Part of the +31 pp headline is attributable to the benchmark’s attribute distribution.* The generator’s abundance of high-EPSS targets amplifies the gap between hygiene-aware and hygiene-blind scoring; under real distributions the aggregate gap is real but small.

(ii) *The relative ordering survives.* HygienePrio-full still tops the leaderboard under real distributions (HP 0.377 $>$

EPSS 0.353 > HRS 0.041 > CVSS 0.014 > Random 0.010). The qualitative claim “hygiene augmentation helps” is preserved; the specific “by 31 pp” magnitude is distribution-dependent.

(iii) *The EEHDA generator can be re-calibrated to real attribute statistics.* Re-fitting the CVE attribute distribution to real EPSS/KEV statistics is a natural next step for benchmark refinement; the corpus and processing pipeline necessary to perform it are released as `real_data/processed/cve_corpus_2020plus.csv` alongside this paper.

(iv) *The mechanism the capacity-decay study names becomes more salient.* The capacity-decay study’s high-capacity collapse mechanism, the high-EPSS tail gets exhausted by capacity-constrained remediation, is substantially more aggressive under the real distribution than under the benchmark one, because the real tail is thinner to begin with. The capacity-decay study’s $K=200$ collapse should therefore appear *earlier* (smaller K) on real distributions, sharpening rather than weakening the capacity-decay study’s central finding.

C. Scope of This Section

This is a partial external-validity test. The real CVE attribute corpus replaces per-CVE attributes only; the fleet topology, host-CVE assignment, HRS components (patch posture, AD exposure, telemetry freshness), and ground-truth definition are still drawn from the generator. A complete external-validity test would require longitudinal exploitation-outcome data from real fleets, which is not available in any public dataset and is documented in §IX as the program’s most consequential remaining gap.

D. Reproducibility

The frozen real CVE corpus, evaluator, and results are released under `real_data/` at the *repository root* (not under `paper4/`):

```
real_data/raw/epss.csv # FIRST.org snapshot 2026-06-05
real_data/raw/kev.json # CISA KEV snapshot
real_data/processed/cve_corpus_2020plus.csv
real_data/processed/manifest.json
real_data/real_dist_evaluator.py
real_data/results/real_dist_results.csv
real_data/results/comparison_summary.json
```

Reproduce with:

```
python3 real_data/real_dist_evaluator.py
```

XI. ADVERSARIAL ROBUSTNESS: A STACKELBERG BEST-RESPONSE ANALYSIS

Vulnerability prioritization scoring is, in principle, a target for strategic attackers: an adversary who knows the defender uses EPSS as a ranking signal can attempt to influence published EPSS scores (through PoC publication timing, social-media-driven exploit chatter, or selective vulnerability disclosure on hosts the attacker has already compromised) to redirect remediation resources. We formalise this scenario as a Stackelberg game and quantify how HygienePrio’s hygiene-augmented scoring compares to EPSS-only under best-response adversarial pressure.

A. Threat Model

A defender deploys a deterministic prioritization scorer S committed in advance. An attacker observes S and chooses a *gaming fraction* $g \in [0, 1]$ representing the share of (host, CVE) pairs whose EPSS attribute the attacker can inflate from its true value to a high-but-plausible target value $\hat{\text{EPSS}} = 0.55$ (approximately the central tendency of EPSS among CVEs that appear in the CISA KEV catalog, mean 0.49, median 0.52 in the frozen corpus of §X, chosen so the gamed pairs become borderline-positive under EPSS-only ranking).

The attacker’s strategic constraint is realism: at high g the defender would notice anomalous EPSS distributions and discount them. We therefore study $g \in \{0, 0.05, 0.10, 0.15, 0.20, 0.30\}$, gaming up to 30% of the eligible pair pool.

The attacker’s strategic preference is to target pairs whose hosts are in the *bottom half* of the HRS distribution. This is the defender’s worst case for HygienePrio: by attacking pairs at low-hygiene-risk hosts, the attacker inflates EPSS in cells where HygienePrio’s HRS co-ranking can do the least to filter them. (If the attacker instead targeted high-HRS hosts, the gamed pairs would co-rank as positives, reinforcing rather than degrading HygienePrio’s top- K selection.)

B. Defender’s Three Strategies

We evaluate the same calibrated HygienePrio scorer weights for HygienePrio-full and the two natural baselines:

- **HygienePrio-full:** scoring weights $(\alpha, \beta, \gamma, \delta) = (0.7, 0.5, 0.1, 0.2)$, unchanged from §5.
- **EPSS-only:** pure exploit-likelihood ranking.
- **HRS-only:** pure hygiene ranking, as a HRS-blind reference that ignores EPSS entirely.

C. Experimental Setup

We run the same 25-seed set as the real-distribution evaluation (105-129; *not* the primary evaluation’s 100-124, see §X) at each of the six gaming levels, yielding 450 result rows. The ground truth (EPSS > 0.10 AND HRS > 75th percentile) is computed *after* attacker gaming, so gamed pairs become spurious positives under EPSS but only true positives if their host happens to clear the HRS threshold, which by construction it does not for attacker-targeted pairs (bottom 50% HRS hosts).

D. Results

Reconciling the $g = 0$ row with Table IV. The unattacked baselines in Table IX (HygienePrio-full 0.595, EPSS-only 0.331) are higher than the primary-evaluation values (0.509 / 0.202) for two identifiable reasons in the released harness (`real_data/adversarial_eval.py`): (a) the adversarial harness restricts the eligible-pair pool to *unpatched* pairs only (`vr[~vr["patched"]]`), whereas the primary evaluation ranks all generated (host, CVE) pairs, removing already-patched pairs raises the positive base rate among ranked candidates and therefore lifts every method’s Precision@50; and (b) the harness uses the 105-129 seed set rather than the

TABLE IX
ADVERSARIAL ROBUSTNESS: MEAN PRECISION@50 ACROSS 25 SEEDS (105-129), BY GAMING FRACTION g (THE SHARE OF LOW-HRS-HOST PAIRS WHOSE EPSS IS INFLATED BY A STACKELBERG ATTACKER). THE $g = 0$ ROW IS THE HARNESS-INTERNAL BASELINE (UNPATCHED-PAIR POOL; SEE §XI) AND IS NOT DIRECTLY COMPARABLE TO TABLE IV.

g	HP-full	EPSS	HRS	HP-EPSS
0.00	0.595	0.331	0.284	+26.4 pp
0.05	0.581	0.195	0.284	+38.6 pp
0.10	0.566	0.194	0.284	+37.3 pp
0.15	0.563	0.194	0.284	+37.0 pp
0.20	0.566	0.193	0.284	+37.3 pp
0.30	0.566	0.193	0.284	+37.4 pp
Drop	-2.9 pp	-13.8 pp	0.0 pp	,

primary 100-124 set. The $g = 0$ row is therefore the harness-internal baseline for the manipulation sweep, not directly comparable to Table IV; all within-sweep deltas are computed against it.

HygienePrio degrades by ≤ 3 pp at any attack level. Across $g \in [0, 0.30]$, HygienePrio-full’s mean Precision@50 falls from 0.595 to 0.566, a -2.9 pp drop at the strongest attack tested. The HRS co-ranking filters gamed pairs because the attacker’s targets have low HRS by construction; HygienePrio’s HRS-component down-weights them despite the inflated EPSS.

EPSS-only degrades by -14 pp at $g = 0.05$. A 5% gaming attack is sufficient to collapse EPSS-only from 0.331 to 0.195 (-13.6 pp) and the drop saturates near -13.8 pp at $g \geq 0.10$. EPSS-only has no host-context information to filter gamed pairs and ranks them in proportion to the inflated EPSS.

The HygienePrio-vs-EPSS gap widens under attack. The benchmark baseline gap is $+26.4$ pp at $g = 0$. At $g = 0.05$ the gap grows to $+38.6$ pp; at $g = 0.30$ to $+37.4$ pp. HygienePrio’s relative advantage over EPSS-only is amplified, not diminished, when an attacker is present.

HRS-only is invariant under EPSS attack. HRS-only’s mean P@50 is 0.284 at every gaming level because HRS-only’s scoring ignores EPSS entirely. This is a methodological sanity check rather than an operational recommendation, HRS-only’s absolute level is below HygienePrio’s at every gaming level studied.

E. Interpretation

The adversarial analysis demonstrates that hygiene augmentation provides a security-relevant property that EPSS-only lacks: *robustness against a strategic adversary who can influence the EPSS signal*. The hygiene side-channel acts as a co-ranking filter that cross-checks exploit-likelihood claims against host-context plausibility. A gamed pair on a low-HRS host is implausibly high-priority and HygienePrio down-ranks it; a gamed pair on a high-HRS host would be a true positive and HygienePrio would correctly elevate it.

This is the canonical defender-advantage pattern in adversarial security: orthogonal signals are robust precisely because the attacker has to game both simultaneously. EPSS+HRS forces the attacker to compromise the hygiene posture of the host

as well as the exploit-likelihood signal, a substantially harder attack than either alone.

F. Caveats

(i) *Benchmark distribution.* This analysis is on the EEHDA benchmark fleet. Section X showed that the benchmark EPSS distribution is heavier-tailed than reality. The adversarial gap may shrink under real distributions for the same mechanism: the EPSS-only baseline starts higher under real distributions, so gaming has less room to inflate. We do not re-run the adversarial sweep under the real distribution; the qualitative finding (HygienePrio is dramatically more robust to EPSS gaming than EPSS-only) is unlikely to reverse because the HRS-cross-check mechanism is distribution-independent.

(ii) *Attacker model is one-shot.* The attacker commits to g before remediation begins; a multi-round adaptive attacker who observes which gamed pairs got remediated could in principle improve. Modelling adaptive attackers requires a different game-theoretic framework and is left as future work.

(iii) *Attack on HRS not modelled.* An attacker who can also manipulate hygiene observables (false patch-state reports, group-membership injection, telemetry-freshness fraud) would undermine HygienePrio’s HRS co-ranking. Modeling adversarial manipulation of host-state telemetry is a substantial open problem; the present analysis bounds the attacker to EPSS manipulation only. The HRS-attack adversary is identified as the strongest plausible threat to hygiene-augmented prioritization and is left as the most important future-work direction in this section.

G. Reproducibility

The adversarial evaluator and frozen results are released under `real_data/` at the repository root (not under `paper4/`):

```
real_data/adversarial_eval.py
real_data/results/adversarial_results.csv (450 rows)
```

Reproduce: `python3 real_data/adversarial_eval.py`.

XII. REAL-WORLD RETROSPECTIVE: WHAT WOULD HYGIENEPRIO HAVE DONE?

The benchmark and real-distribution evaluations of §§VII to XI test HygienePrio on distributional aggregates. We close the empirical evaluation with a *worked retrospective*: for ten high-profile CVEs of the past seven years, we examine what HygienePrio would have produced relative to EPSS-only at the moment of public disclosure, using the present-day (2026-06-05) EPSS and KEV state as a stable late-life reference.

A. Observations

(i) **EPSS catches major CVEs.** All ten high-profile CVEs have present-day EPSS scores in $[0.9342, 0.9446]$, all within the top 0.2% of the EPSS distribution (per-CVE percentiles range from 99.825 to 99.996). Every one is in the CISA KEV catalog. EPSS-only ranking would correctly elevate any of these CVEs to the top of a fleet’s prioritization queue. The

TABLE X
REAL CVE RETROSPECTIVE. TEN HIGH-PROFILE CVEs FROM 2014-2024, WITH
PRESENT-DAY (2026-06-05) EPSS SCORES AND KEV INCLUSION DATES. SOURCE:
FIRST.ORG EPSS FEED + CISA KEV CATALOG.

CVE	EPSS	KEV date	Name
CVE-2021-44228	0.944	2021-12-10	Log4Shell
CVE-2014-0160	0.945	2022-05-04	Heartbleed
CVE-2023-23397	0.934	2023-03-14	Outlook NTLM
CVE-2023-34362	0.943	2023-06-02	MOVEit SQLi
CVE-2023-4966	0.943	2023-10-18	Citrix Bleed
CVE-2024-3400	0.943	2024-04-12	PAN-OS RCE
CVE-2022-30190	0.936	2022-06-14	Follina MSDT
CVE-2021-34527	0.942	2021-11-03	PrintNightmare
CVE-2020-1472	0.944	2021-11-03	Zerologon
CVE-2019-0708	0.945	2021-11-03	BlueKeep

*All ten in top 0.2% of EPSS distribution.
EPSS-only cannot discriminate among them.*

first-order claim “EPSS works for major CVEs” is empirically supported by the present-day data.

(ii) **Major CVEs leave EPSS no room to discriminate.** The narrow EPSS range [0.9342, 0.9446] across ten distinct high-impact CVEs means EPSS-only’s relative ranking among them is essentially noise. A 2023 fleet on the day MOVEit Transfer was disclosed (CVE-2023-34362, EPSS = 0.9425, KEV-added 2023-06-02) faced the question: *which of our hosts running MOVEit should we patch first?* EPSS-only provides no answer, every affected host has the same per-CVE score.

(iii) **HygienePrio’s contribution is host-side discrimination.** The HygienePrio scorer’s EPSS·HRS interaction term $\delta \cdot (\text{EPSS}(c) \cdot \text{HRS}(h))$ multiplies the high EPSS of a major CVE by the host’s hygiene risk, breaking the tie in favour of high-HRS hosts (high patch debt, high AD exposure, stale telemetry). The HRS dimension that the HygienePrio scorer §VII evaluation identifies as most informative, *patch posture*, is exactly the dimension that captures which hosts are likely to have a long-unpatched MOVEit instance. On a retrospective Log4Shell-style fleet sweep with thousands of hosts running affected software, HygienePrio’s ranking would elevate the patch-debt-laden hosts first; EPSS-only would return them in arbitrary order.

(iv) **The KEV signal arrives after disclosure.** For CVEs like Log4Shell (disclosed 2021-12-09, KEV-added 2021-12-10), KEV inclusion lags disclosure by one day; for Heartbleed (disclosed 2014-04-07, KEV-added 2022-05-04) the KEV-added date is over *eight years* after disclosure, a function of the KEV catalog’s 2021 inception, not of CVE risk. The HygienePrio $\gamma \cdot \text{KEV_recency}(c)$ term contributes meaningfully in the days after KEV inclusion; before KEV inclusion it does not fire. EPSS scores, by contrast, are updated daily for every CVE regardless of KEV status. The KEV term in HygienePrio is therefore a *lagging amplifier* of CVEs that EPSS has already flagged.

B. The Realistic Operational Claim

The empirical results across this paper, including the §X real-distribution finding, sharpen the +31 pp benchmark headline into a precise operational claim:

HygienePrio provides host-side discrimination among the high-EPSS CVEs that EPSS-only treats as interchangeable. Under the real-world distribution of CVE attributes, the relative advantage of HygienePrio over EPSS-only is small in aggregate Precision@50 (+2.4 pp, §X), but is operationally significant at the moment of a major-CVE disclosure: it tells a remediation team which of many vulnerable hosts to patch first.

This is the claim we recommend a deploying organisation use as the basis for selecting HygienePrio over EPSS-only. The +31 pp benchmark gap is a faithful measurement on the EEHDA fleet model; the aggregate real-distribution benefit is smaller, and the host-side discrimination value at major-CVE moments is the operationally decisive rationale.

C. Caveats

(i) *Late-life EPSS proxy for disclosure-time EPSS.* The EPSS scores in Table X are present-day (2026-06-05) values, not the disclosure-time values. EPSS scores evolve over time as exploitation evidence accumulates. The direction of evolution for major CVEs is upward (toward 1.0 as they receive widespread attention), so the disclosure-day EPSS for these CVEs was generally lower than the present-day value. We use the present-day value as a proxy because the FIRST.org public feed provides current scores; historical EPSS trajectories require the private FIRST historical archive.

(ii) *Retrospective is anecdote, not evaluation.* This subsection presents ten anecdotes about real CVEs to motivate the operational claim. It is not a controlled evaluation. The aggregate-distribution evaluation in §X is the proper external-validity test.

D. Reproducibility

`real_data/processed/famous_cves_snapshot.csv`

(10 CVEs with present-day EPSS, KEV date, and severity description.) Built from the FIRST.org and CISA snapshots in `real_data/raw/`.

XIII. AN INFORMATION-THEORETIC CEILING ON HYGIENE AUGMENTATION

The empirical gap between the benchmark (+30.9 pp) and real-distribution (+2.4 pp) Precision@50 improvements (§X) invites a question with a definite mathematical character: *how much could hygiene augmentation possibly help, given the information each signal carries about the labelling target?* We frame this as a mutual-information bound and apply it to whatever subset of the real corpus permits direct estimation.

A. Bound

Let the ground-truth label at a (host, CVE) pair be $Y \in \{0, 1\}$. Let X_E denote the EPSS score and X_H the HRS score, viewed as random variables. The Bayes-optimal achievable top- K Precision from a feature set $\mathbf{X}$ is bounded above by a monotone function of the mutual information $I(Y; \mathbf{X})$ via

Fano’s inequality [13]. Crucially, the marginal benefit of adding HRS to EPSS is bounded by the conditional mutual information:

$$P_K^*(X_E, X_H) - P_K^*(X_E) \leq f(I(Y; X_H | X_E)), \quad (4)$$

for some monotone increasing $f(\cdot)$ that depends on K and on the positive-mass $p = \mathbb{P}[Y = 1]$. *If HRS carries no information about Y that EPSS does not already carry, the marginal benefit of adding HRS is zero*, regardless of how the two signals are combined in the scorer.

B. Estimation on Available Data

We can estimate $I(Y; X_E)$ directly from the real CVE corpus (§X) using KEV inclusion as the labelling proxy $Y = \mathbf{1}[\text{CVE} \in \text{KEV}]$. The plug-in estimator with five-bin discretisation of EPSS on the 203,174-CVE corpus yields

$$\hat{I}(Y; X_E) \approx 0.010 \text{ nats} \quad (5)$$

(≈ 0.014 bits), consistent with the small KEV prevalence (0.52%) producing low total entropy.

The estimation of $I(Y; X_H | X_E)$ on real data is not directly possible because *the HRS feature does not exist in any public CVE feed*. HRS combines patch-posture, AD-exposure, and telemetry-freshness signals that are properties of *the defender’s fleet*, not of the CVE record. The publicly observable analogue, the EPSS percentile rank of CVEs known to affect manageability-relevant software, correlates strongly with EPSS itself and contributes negligible conditional information ($\hat{I}(Y; X_H^{\text{proxy}} | X_E) \approx 0.0003$ nats on the real corpus). The actual conditional MI of a true HRS signal with respect to a real-fleet exploitation outcome could be substantially higher, but estimating it requires real-fleet telemetry that is unavailable publicly.

C. What the Bound Argument Says

Without a real-data MI estimate for $I(Y; X_H | X_E)$ we cannot compute the real-data ceiling directly. We can, however, make three observations bounded by the inequalities (4):

(i) *The real-distribution gap is consistent with low conditional information.* The empirical +2.4 pp real-distribution gap (§X) is achievable from a small $I(Y; X_H | X_E)$. The fact that the observed gap is small under real distributions does not by itself prove HRS carries little conditional information; it is consistent with both “HRS carries little CMI” and “HRS carries CMI but HygienePrio’s linear scorer does not extract it efficiently.”

(ii) *The benchmark gap exceeds any plausible real-data MI budget.* The EEHDA fleet model couples HRS into the ground-truth predicate by construction (the labelling rule is $\text{EPSS} > 0.10$ AND $\text{HRS} > 75\text{th percentile}$). This makes $I(Y_{\text{bench}}; X_H | X_E)$ structurally large by design. The benchmark ceiling is therefore an upper bound on what hygiene augmentation could achieve *under the benchmark labelling rule*, not on what it would achieve under real exploitation outcomes.

(iii) *The major-CVE retrospective is a different conditional regime.* Section XII operates on the conditional $P(Y | X_E = \text{near-1.0})$ where EPSS provides essentially no discriminative

information (every major CVE has top-percentile EPSS). In that conditional regime, the relevant quantity is $I(Y; X_H | X_E = \text{high})$, which is plausibly larger than the marginal $I(Y; X_H | X_E)$ averaged over the whole EPSS range. The information-theoretic framing supports the §XII operational claim more strongly than it supports the aggregate-distribution claim.

D. Conclusion of the Bound Argument

A precise information-theoretic ceiling computation requires real-fleet exploitation-outcome data to estimate $I(Y; X_H | X_E)$. That data does not exist publicly. The most we can say from the public corpus is:

- The marginal information EPSS carries about KEV inclusion is small in absolute terms (≈ 0.014 bits), bounding any Precision@K improvement at modest levels even at the Bayes-optimal limit.
- The benchmark Precision@K gaps observed for the HygienePrio scorer exceed what real-data information budgets plausibly support, consistent with the distribution-dependent component of the headline identified in §X.
- The information-theoretic ceiling cannot be computed precisely without a real HRS signal correlated with real exploitation outcomes. This is the same gap identified in §XII: the program’s most consequential open problem is access to real fleet exploitation-outcome data, not algorithm refinement.

E. Caveats on the Estimates

(i) *The MI estimates here are noisy plug-in estimates with five-bin uniform discretisation; finer bins and bias-corrected estimators would change the magnitudes by $\pm 50\%$ but not the qualitative conclusion.*

(ii) *KEV inclusion is a delayed, conservative labelling proxy.* KEV-listed CVEs are those CISA has confirmed actively exploited. The “true” positive set for vulnerability prioritization is broader (includes CVEs that will be exploited but have not yet been), and that broader set is unobservable at any given moment.

(iii) *HRS proxy is structural, not predictive.* The HRS-substitute used in the conditional MI estimate is a constructed noisy correlate of Y rather than a real-fleet HRS measurement. The reported $\hat{I}(Y; X_H^{\text{proxy}} | X_E) \approx 0.0003$ nats is the floor that public data allows us to verify; the ceiling is unknown.

XIV. RELATED WORK

This section situates HygienePrio within the broader literature on vulnerability management, host-context scoring, and cyber-hygiene measurement, supplementing the foundational background in §2.

Vulnerability scoring and exploit prediction. The Common Vulnerability Scoring System (CVSS) [14] provides a standardised framework for assessing severity across Base, Temporal, and Environmental metrics. As a CVE-level signal, CVSS shares the asset-agnostic limitation of EPSS [1]: it scores a vulnerability in isolation without regard to the

maintenance state of the affected host. Jacobs et al. [2] demonstrated that exploit-prediction models substantially outperform CVSS-based triage for reducing mean time to remediation on enterprise data. This foundational result motivates using EPSS as the primary exploit-likelihood signal in HygienePrio. The National Vulnerability Database (NVD) [15] serves as the primary aggregation point for CVSS scores and CVE metadata. Unlike the above work, which optimises exploit prediction as a standalone CVE-level signal, HygienePrio integrates exploit likelihood with host-level hygiene posture via an explicit interaction term, targeting the (host, CVE) pair rather than the CVE alone.

Risk-based vulnerability management (RBVM). Commercial RBVM platforms, Tenable Lumin, Qualys TruRisk, and Rapid7 InsightVM, combine EPSS, CVSS, threat intelligence, and asset context in proprietary scoring models that address the same host-CVE pair prioritization problem as HygienePrio. These systems are closed and not independently reproducible; no comparison claims are made against them. The 2026 Verizon Data Breach Investigations Report [10] documented that vulnerability exploitation surpassed stolen credentials as the leading breach entry point for the first time, underscoring the operational urgency of principled patch prioritization at scale.

Capacity-constrained prioritization. The VulnPrio study [5] constructed a reproducible benchmark for (host, CVE) pair prioritization on a synthetic government-shaped fleet, evaluating thirteen strategies combining EPSS, CVSS, KEV membership, asset criticality, and endpoint exposure across thirty seeds. Its central finding was null: no CVE-level augmentation materially outperformed EPSS, attributing this to uncalibrated weights and the absence of host-level hygiene signals. HygienePrio directly addresses this gap by introducing a three-dimensional Hygiene Risk Score.

Identity and access management (IAM) security. The foundational concepts of access control and identity state that underpin the Active Directory (AD) Exposure dimension of HRS are surveyed in Bertino & Sandhu [16]. AD-exposure-aware risk scoring has been proposed in the lateral movement risk literature; HygienePrio incorporates AD exposure as one of three HRS dimensions, operationalizing domain-privilege exposure as a host-level risk amplifier rather than as a standalone signal.

Anomaly detection for hygiene monitoring. Aggarwal & Sathe [8] establish theoretical foundations for outlier ensemble methods applicable to cyber-hygiene anomaly detection. HygieneBench [6] applied eight such methods across seven hygiene monitoring tasks on a synthetic fleet, finding that rule baselines dominate 86.2% of configurations but that patch-vulnerability hygiene (Task T5) and group-membership drift (Task T2) are exceptions where structured methods add meaningful signal. HygienePrio operationalises these two discriminative dimensions, patch posture and AD exposure, as the dominant components of HRS, grounded in HygieneBench’s empirical findings.

Telemetry freshness and policy mandates. CISA BOD 23-01 [4] mandates 14-day asset discovery and 72-hour vulnerability data freshness for federal agencies, and NIST

SP 800-53 [17] specifies continuous monitoring controls that depend on current telemetry. These policies encode the intuition that stale telemetry degrades risk assessment reliability. HygienePrio models telemetry freshness explicitly as the TelemetryFreshness component of HRS, formalizing the policy mandate as a quantitative host-level risk signal.

Weight calibration methodology. The pre-registered evaluation design, seed-stratified BCa bootstrap confidence intervals [11], and capacity-constrained scheduler are inherited from the VulnPrio study [5]; the calibration gate methodology is inherited from our earlier calibration analysis on failure-aware public-feed gating. The threats taxonomy follows Wohlin et al. [12].

No prior work combines exploit-likelihood scoring with a multi-dimensional, telemetry-derived host hygiene score and an explicit interaction term in a pre-registered, reproducible benchmark. HygienePrio is the first system to jointly operationalise patch posture, AD exposure, and telemetry freshness as a unified HRS integrated with EPSS for (host, CVE) pair prioritization.

XV. CONCLUSION

The central finding of this paper is that host-level hygiene posture, patch debt, AD privilege exposure, and telemetry freshness, adds substantial prioritization value over EPSS-only rankings when operationalized as a Hygiene Risk Score and integrated into a pre-calibrated weighted scorer. On a pre-registered, fully reproducible benchmark over the EEHDA fleet, HygienePrio-full outperforms EPSS-only by 31 pp at P@50 across all 25 of 25 evaluation seeds (non-overlapping BCa 95% CIs). Under real EPSS/KEV attribute distributions the aggregate gap is +2.4 pp (§X), while host-side discrimination at major-CVE disclosure moments, exactly when remediation capacity is scarcest, remains the operationally significant advantage (§XII). The practical take-home message is concrete: of the three hygiene dimensions, patch posture alone accounts for ≈ 20 pp of this improvement. Operations teams with limited instrumentation capacity should prioritize patch posture tracking above more complex hygiene signals such as AD group exposure or telemetry freshness monitoring.

This result completes a three-paper methodological arc. The VulnPrio study (VulnPrio) established that CVE-level feature augmentation does not improve over EPSS-only, explicitly attributing the null to the absence of dynamic host-level signals. The HygieneBench benchmark (HygieneBench) demonstrated that patch posture and AD group drift carry discriminative structure in fleet telemetry. HygienePrio operationalizes these signals in a remediation prioritization scorer. The progression from honest null through diagnostic signal identification to positive integration result illustrates how pre-registered negative results generate scientific value by constraining and directing subsequent work.

The headline magnitudes are benchmark measurements. The ground-truth definition incorporates HRS as a labelling condition, which contributes a structural component to the benchmark gap that real-exploitation-data evaluation would remove; the external-validity and retrospective analyses (§§X, XII)

quantify and bound this component. The calibrated weights were estimated on five seeds and should be re-fitted before real-fleet deployment. We therefore present HygienePrio as a validated benchmark contribution with a characterized boundary of applicability, ahead of prospective real-fleet evaluation.

Reproducibility. The evaluation code, fleet generator, frozen results, and pre-registration protocol are available in the accompanying repository at <https://github.com/Harshavardhanmalla-Labs/research-papers/tree/main/paper4>. The real-data artifacts referenced in §§X to XII live under `real_data/` at the repository root (not under `paper4/`).

Future work. The most important next step is evaluation on real, appropriately anonymized fleet telemetry using exploitation event logs (not HRS) as ground truth. Secondary directions include: (i) extending HRS with network-layer signals (firewall coverage, internet-facing service flags) to test whether additional dimensions contribute beyond the current three; (ii) evaluating HygienePrio under rolling daily EPSS score updates rather than static per-seed snapshots; and (iii) testing whether non-linear scoring functions trained on historical remediation outcomes can outperform the pre-calibrated linear scorer, which is the binding simplification assumption of this work.

DATA AVAILABILITY STATEMENT

The data and code that support the findings of this study are openly available in the research-papers repository at <https://github.com/Harshavardhanmalla-Labs/research-papers/tree/main/paper4>. The manuscript sources, figures, analysis code, and frozen result tables are included and regenerate every reported number deterministically from the fixed seeds.

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